Dawei Cai

Electrical and Computer Engineering

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Education

University of Washington — Seattle, WA
B.S. Electrical and Computer Engineering — *Expected June 2027*

Work Experience

Yilin Clothing Co., Ltd. — June 2022 – Sep 2022
Supply Chain Intern

- Analyzed supply chain data including sales, inventory, and production records
- Identified operational inefficiencies such as inventory overstock and delays in procurement cycles
- Assisted in generating reports to support data-driven decision making and process improvements

Project Experience

FPGA Digital Systems Design Suite — Jan 2026 – present
SystemVerilog, Quartus, ModelSim, DE1-SoC

- Designed and implemented multiple digital systems on FPGA including search accelerators, VGA graphics, and audio processing modules
- Developed FSM-based control logic and datapath architectures for hardware-accelerated binary search
- Implemented VGA graphics pipeline using Bresenham's line algorithm with framebuffer-based pixel generation
- Built audio processing system with microphone loopback, tone generation, and FIR filtering using synchronous handshake protocols
- Verified system behavior through ModelSim waveform simulation and FPGA hardware testing

Embedded Line-Following Robotic System — Jan 2026 – present
Arduino, PCB Design, Fusion360

- Designed and built an autonomous line-following robot integrating sensors, motor drivers, and embedded control logic
- Developed real-time control algorithms for path tracking using sensor feedback
- Designed custom PCB and performed circuit soldering for hardware integration
- Created mechanical chassis using Fusion360 and implemented 3D-printed components
- Integrated hardware and software subsystems to achieve stable real-time operation

Activities

The Fifth Annual Conference of China Robotics Society — Aug 2024
Volunteer and Participant

- Attended technical sessions to learn about current developments and research directions in robotics
- Assisted graduate and PhD students in demonstrating robotics systems and interactive exhibits
- Gained exposure to practical robotics applications in industry, healthcare, and service systems

Smile Angel Foundation — Jan 2022 – Jun 2023
Volunteer

- Collected and organized information for infants eligible for cleft lip and palate surgery support programs
- Used Excel spreadsheets to assist communication and data coordination between medical staff and patient families
- Supported charity project operations including information tracking and resource coordination

Technical Skills

Programming: Java (OOP), Python, SystemVerilog
Embedded : FPGA (DE1-SoC), Arduino, Digital Logic Design, FSM, Datapath Design
Tools : Quartus, ModelSim, Git, Fusion360, AutoCAD
Other : PCB Design, Circuit Soldering, Sensor Integration, Waveform Debugging